Computer Science

- Pick a Language
  - Python
  - Go
  - C#
  - C++
  - C
  - Java
  - Rust
- Data Structures
  - Array
  - Linked List
  - Stack
  - Queue
  - Tree
  - Hash Table
  - Binary Tree
    - Binary Search Tree
    - Full Binary Tree
    - Complete Binary Tree
    - Balanced Tree
    - Unbalanced Tree
  - Graph
    - Directed Graph
    - Undirected Graph
    - Spanning Tree
    - Representation
      - Adjacency List
      - Adjacency Matrix
  - Heap
- Asymptotic Notation
  - Big O
    - Big-Theta
    - Big Omega
  - Common Runtimes
    - Constant
    - Logarithmic
    - Linear
    - Polynomial
    - Exponential
    - Factorial
- Common Algorithms
  - Graphs
    - Breadth First Search
    - Depth First Search
    - Bellman Ford's Algorithm
    - Dijkstra's Algorithm
    - A* Algorithm
  - Tree
    - Pre-Order Traversal
    - In-Order Traversal
    - Post Order Traversal
    - Breadth First Search
    - Depth First Search
  - Recursion
    - Tail Recursion
    - Non-Tail Recursion
  - Searching
    - Binary Search
    - Linear Search
  - Sorting
    - Bubble Sort
    - Selection Sort
    - Insertion Sort
    - Heap Sort
    - Quick Sort
    - Merge Sort
  - Greedy Algorithms
    - Dijkstra's Algorithm
    - Huffman Coding
    - Kruskal's Algorithm
    - Ford-Fulkerson Algorithm
    - Prim's Algorithm
  - Back Tracking
    - Finding Hamiltonian Paths
    - Solving N Queen Problem
    - Maze Solving Problem
    - The Knight's Tour Problem
    - Rabin-Karp Algorithm
  - Caches
    - MFU Cache
    - LRU Cache
    - LFU Cache
- String Search & Manipulations
  - Search Pattern in Text
  - Suffix Arrays
  - Substring Search
    - Brute Force Search
    - Robin-Karp
    - Knuth-Morris Pratt
    - Boyer-Moore
- Floating Point Math
- Endianness
  - Big Endian
  - Little Endian
- Character Encodings
  - Unicode
  - ASCII
- Bitwise Operators
- Common UML Diagrams
  - Class Diagrams
  - Usecase Diagrams
  - Activity Diagrams
  - Statemachine Diagrams
  - Sequence Diagrams
- Design Patterns
  - GoF Design Patterns
  - Architectural Patterns
  - Dependency Injection
  - Null Object Pattern
  - Type Object Pattern
- Basic Math Skills
  - Probability
  - Combinatorics
- Complexity Classes
  - P
  - NP
  - P = NP
  - Co-NP
  - NP Hard
  - NP Complete
    - Travelling Salesman Problem
    - Knapsack Problem
    - Longest Path Problem
- Tries
- Networking
  - OSI Model
  - TCP/IP Model
  - DNS
  - HTTP
  - TLS & HTTPS
  - Sockets
- Databases
  - SQL vs NoSQL Databases
  - Normalization / Denormalization
  - Entity-Relationship Model
  - DDL
  - DML
  - DQL
  - DCL
  - Locking
  - ACID Model
  - BASE
  - CAP Theorem
  - PACELC
  - Indexes
  - Views
  - Transactions
  - Stored Procedures
  - Database Federation
  - Replication
  - Sharding
- System Design
  - Horizontal vs Vertical Scaling
  - Load Balancing
  - Clustering
  - Caching
  - CDN
  - Proxy
  - CAP Theorem
  - Queues
  - Architectural Styles
    - REST
    - GraphQL
    - gRPC
  - Cloud Design Patterns
  - Long Polling
  - Short Polling
  - Web Sockets
  - SSE
- Balanced Search Trees
  - AVL Trees
  - Red / Black Trees
  - 2 3 Search Trees
  - 2 3 4 Trees
  - K-ary / M-ary Tree
  - B-Tree
  - K-D Trees
  - Skip Lists
- Security
  - Hashing / Encryption / Encoding
  - Public Key Cryptography
  - Hashing Algorithms
  - OWASP Top 10
- Processes and Threads
  - Process Forking
  - Memory Management
  - Lock / Mutex / Semaphore
  - Concurrency in Multiple Cores
  - Scheduling Algorithms
  - CPU Interrupts
  - Processes vs Threads
- How Computers Work
  - How CPU Executes Programs
  - Registers and RAM
  - Instructions and Programs
  - CPU Cache
  - How Computers Calculate